

Abhishek Rout

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Research Interests

Theoretical and numerical gravitational physics; modified gravity (Chern-Simons, parity-violating extensions); gravitational perturbation theory in black hole and cosmological spacetimes; quantum field theory in curved backgrounds; gravitational wave physics; early universe cosmology.

Education

Ph.D. in Physics, University of South Carolina, Columbia, SC 2019–2025

Awarded: December 15, 2025 — GPA: 3.817/4.00

Advisor: **Prof. Brett Altschul**

Dissertation: *Gravitational Perturbations and Quantum Field Interactions in Modified Spacetime Backgrounds*

Integrated BS-MS in Physics, IISER Kolkata, India 2014–2019

CGPA: 8.74 / 10 (Top 10% of cohort)

Doctoral Research Experience

University of South Carolina 2019 – 2025

- Developed novel analytical and numerical methods to study tensor and metric perturbations in black hole and cosmological space-times.
- Investigated parity-violating extensions of general relativity, with focus on Chern-Simons modified gravity and its observational signatures.
- Created computational pipelines in Python and Mathematica to solve coupled field equations in curved spacetime, implementing numerical ODE/PDE solvers and symbolic computation.
- Discovered novel bound state formations and particle production mechanisms in breather-type field configurations produced by solitons.
- Derived predictions for primordial gravitational wave birefringence and black hole perturbations, leading to three peer-reviewed publications.

Technical & Computational Skills

Programming: Python, C, MATLAB, Mathematica, $\text{\LaTeX}$

Scientific Computing: Numerical relativity techniques; ODE/PDE solvers; symbolic algebra; Linux/Windows environments.

Physics Tools: Tensor calculus; perturbation theory; cosmological perturbation theory; black hole perturbation theory; field theory in curved spacetime.

Publications

Peer-Reviewed

1. A. Rout and B. Altschul, “Bound States and Particle Production by Breather-Type Background Field Configurations,” *Symmetry* **16**, 1571 (2024),
DOI:10.3390/sym16121571.
2. A. Rout and B. Altschul, “Primordial Gravitational Wave Birefringence in a de Sitter Background with Chern–Simons Coupling,” *Universe* **12**, 6 (2026),
DOI:10.3390/universe12010006.
3. A. Rout and B. Altschul, “First-Order Axial Perturbation of the Reissner–Nordström Metric In a Possible Chern–Simons Gravity Background,” *Symmetry* **18**, 560 (2026)
DOI:10.3390/sym18040560.

Preprints

1. A. Rout and B. Altschul, “Gravitational Wave Parity-Violating Strain from Pulsar Glitches in Chern–Simons Modified Gravity,” *arxiv* preprint(Submitted to PRD),
DOI:arXiv:2608.24002.
2. A. Rout and B. Altschul, “Mixed-Order Parity-Violating Corrections to Rotating Black Holes in Combined Scalar–Gauss–Bonnet and Dynamical Chern–Simons Gravity,” *In preparation*
3. A. Rout and B. Altschul, “Isospectrality breaking in the axial quasinormal modes of Reissner–Nordström black holes in dynamical Chern–Simons Gravity,” *In preparation*
4. A. Rout and B. Altschul, “Rotation-induced isospectrality breaking of charged black holes in dynamical Chern–Simons Gravity,” *In preparation*

Pre-Doctoral Research Experience

Master’s Research Project, IISER Kolkata 2018 – 2019

- Theoretical investigation of gravitational collapse in spherically symmetric spacetimes using analytical methods.
- Analyzed scalar-field-driven collapse for quadratic potentials in dust-dominated cosmological models.

Advanced Theoretical Project, IISER Kolkata 2017

- Developed computational framework in Python to study the twin paradox in Schwarzschild spacetime.
- Examined inertial and accelerated trajectories near compact objects, comparing numerical and analytical results.

Undergraduate Research, IIST Trivandrum 2017

- Studied foundational aspects of loop quantum gravity and spin network states.
- Explored applications of entangled loop quantum states in quantum optics protocols.

Undergraduate Research, IISER Kolkata 2015 – 2016

- Computational studies in special and general relativity using MATLAB and Python.
- Modeled relativistic effects including the perihelion precession of Mercury, achieving $< 1\%$ numerical error.

Presentations & Workshops

- **Contributed Talk**, Indiana University Center for Spacetime Symmetries(IUCSS) Workshop on Theoretical Aspects of the Standard-Model Extension and Modified Gravity (2024).

- **Attendee**, Indiana University Center for Spacetime Symmetries(IUCSS) Workshop on Gravitational Aspects of Lorentz Violation (2023).
- **Seminar Speaker**, USC Theoretical Physics Group Seminar Series (2023, 2024 and 2025).

Fellowships and Honors

- **INSPIRE Fellow**, Department of Science and Technology (DST), Government of India (2014–2019).
- Selected for **DST Science Awareness Camp**, NIST Berhampur (top 50 students nationally).
- Ranked in the **top 1%** in CISCE Board Examinations 2014.

Teaching & Outreach

Teaching Assistant, Department of Physics & Astronomy, University of South Carolina 2019–2025

- **General Physics I (PHYS 201: Newtonian Mechanics)**
 - Led weekly discussion sections for ~ 40 students, clarifying concepts in kinematics, dynamics, and conservation laws.
 - Developed and graded homework assignments, quizzes, and exam problems; provided detailed feedback.
- **General Physics II (PHYS 202: Introduction to Electromagnetism)**
 - Assisted in laboratory setup and execution, guiding students through experiments on electric fields, circuits, and magnetism.
 - Created tutorial materials and visual aids (using Python simulations) to demonstrate complex electromagnetic phenomena.

Science Outreach Volunteer, IISER Kolkata and Local Schools

2015–2019

- Designed and delivered interactive physics demonstration workshops for high school students.
- Topics included principles of mechanics, wave optics, and astronomy using hands-on models and simulations.

Professional Activities

- **Co-referee** for journal articles under the supervision of Prof. Altschul on the widespread topics of *Classical and Quantum Gravity*.
- **Member**, Music Club and Badminton Club, IISER Kolkata.

References

- **Prof. Brett Altschul** (Advisor), Professor, Department of Physics and Astronomy, University of South Carolina. Email: altschul@mailbox.sc.edu
- **Prof. Matthias Schindler** (Additional Reference), Professor, Department of Physics and Astronomy, University of South Carolina. Email: mschindl@mailbox.sc.edu
- **Prof. Richard Creswick** (Additional Reference), Professor, Department of Physics and Astronomy, University of South Carolina. Email: creswick@mailbox.sc.edu

Complete references and contact information available upon request.